Language Agent Tree Search

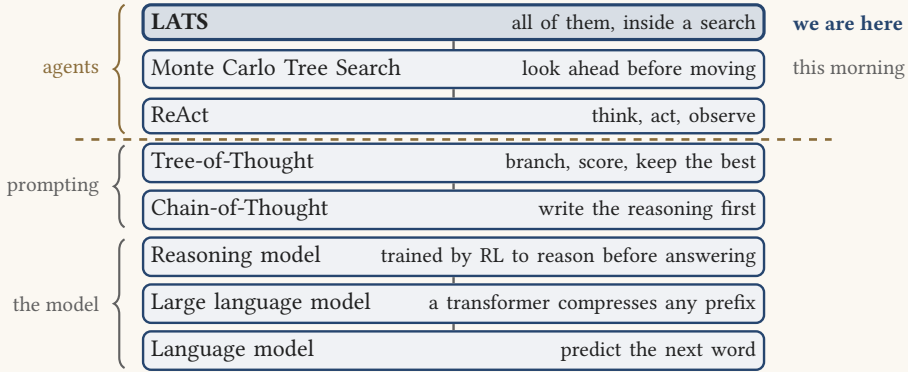
One frozen model as a planning agent

Carlos Cotrini

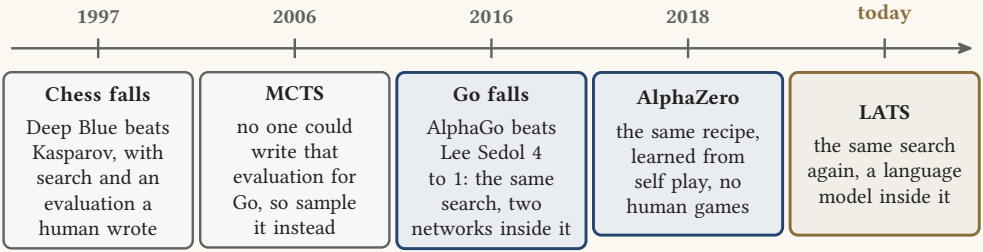
ETH Zurich

21 October 2026

Where LATS sits



Monte Carlo Tree Search, before language models



The search itself barely changed across twenty years. What changed each time is what supplies the two things it needs: something to propose moves, and something to judge a position. LATS supplies both from one frozen model.

Coulom 2006; Kocsis and Szepesvari 2006; Silver et al., Nature 2016; Silver et al., Science 2018.

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Section 1

The LATS method

UCT: the critic, plus an exploration bonus

$$\mathrm{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w\sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

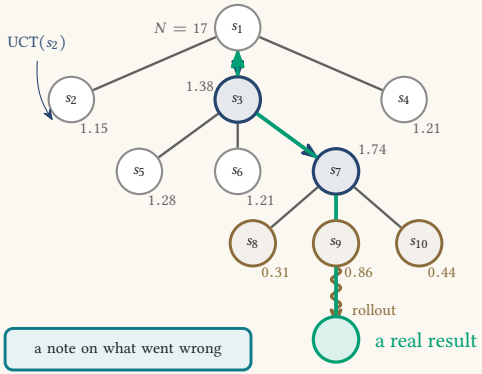
The critic
what the model thinks this branch is worth

+

The exploration bonus
how little we have tried it

This morning's words for the same two terms: how good it looked, plus how little we know.

The same loop as this morning, with four edits



1 Selection by UCT

2 Expansion the *actor* proposes

3 Evaluation the *critic* scores

4 Simulation the *best* one, to the end

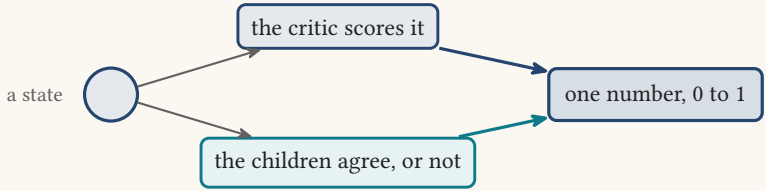
5 Backpropagation the *real* result, up

6 Reflection the *feedback* model writes it

into the next prompt

grey: UCT, what selection compares.
amber: V from the critic, and all that is new.

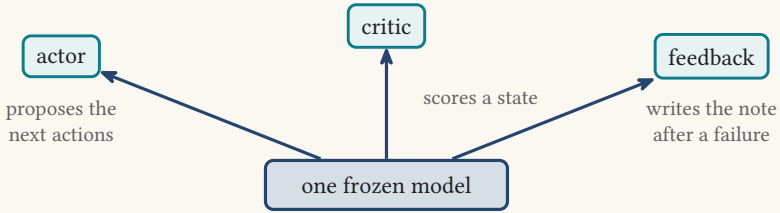
What the critic actually returns



$$V(s) \;=\; \lambda\,\mathrm{LM}(s) \;+\; (1-\lambda)\,\mathrm{SC}(s)$$

$\lambda = 0.5$ on question answering, 0.8 on programming and web tasks.

Three jobs, one set of weights



same weights every time. Only the prompt changes.